

AlgoNexus: A Revised IEEE-Format Report on a Web-Based Recursion Tree Visualizer for Divide-and-Conquer Algorithms

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Abstract—Understanding recursive algorithms is difficult because students must mentally track nested calls, sub-problem decomposition, return propagation, and the relationship between a recurrence and its concrete execution behavior. This revised report presents *AlgoNexus*, a browser-based educational system designed to visualize divide-and-conquer algorithms as interactive recursion trees. The platform supports Merge Sort, Quick Sort, Binary Search, Closest Pair of Points, and Strassen’s Matrix Multiplication, thereby exposing learners to balanced binary recursion, logarithmic single-branch recursion, geometric divide-and-conquer, and higher branching-factor recursion within one interface. The system combines a modular three-layer architecture with a state manager, step-wise navigation controls, contextual annotations, and color-coded node states that distinguish active, visited, and pending calls. In contrast to many general-purpose algorithm animation tools, the emphasis here is not only on showing values but also on explaining the semantic purpose of each recursive step. This IEEE-style revision strengthens the original report by adding formal figures, comparative tables, explicit design rationale, and a deeper discussion of visualization methodology, implementation trade-offs, and future extensibility. The resulting document is intended as a clearer and more publication-style presentation of the first project phase.

Index Terms—algorithm visualization, divide and conquer, recursion tree, computer science education, web-based learning tool, Strassen matrix multiplication

I. INTRODUCTION

Recursion is central to undergraduate algorithm design, yet it remains one of the most conceptually demanding topics for students. A learner may understand a recurrence such as $T(n) = 2T(n/2) + O(n)$ algebraically while still lacking intuition about how recursive calls expand, how depth evolves, why subproblems appear in a particular order, and how intermediate results combine during return. This gap between notation and process is especially visible in divide-and-conquer algorithms, where the quality of understanding depends on seeing the *structure* of decomposition rather than merely memorizing pseudocode.

AlgoNexus was developed to address this gap through a browser-accessible recursion-tree visualizer. The system renders recursive execution as a navigable sequence of tree states, each accompanied by contextual explanation. Instead of treating the recursion tree as an abstract object discussed after the fact, the interface allows the learner to observe it emerge step by step during execution. This design choice aligns with

prior findings that algorithm visualization is most effective when learners actively explore rather than passively watch animations [1].

The revised report retains the original academic purpose of the project while reorganizing the content into a more formal IEEE-style structure. The main improvements over the source version are threefold. First, figure placeholders have been replaced with concrete diagrams and screenshots. Second, the narrative has been strengthened with comparative discussion, architectural formalization, and explicit mapping between theory and visualization behavior. Third, the document now frames the platform more clearly as an educational system whose contribution lies in the combination of algorithm coverage, interaction design, and explanatory scaffolding.

A. Project Scope

The current implementation covers five divide-and-conquer algorithms chosen to expose distinctly different recursive patterns:

- **Merge Sort:** balanced binary recursion with explicit combine phase;
- **Quick Sort:** partition-driven recursion whose depth depends on pivot balance;
- **Binary Search:** single-branch logarithmic recursion;
- **Closest Pair of Points:** geometric divide-and-conquer with cross-boundary reasoning; and
- **Strassen’s Matrix Multiplication:** seven-subproblem matrix recursion with sub-cubic complexity.

These algorithms collectively allow students to compare branching factor, recursion depth, and asymptotic behavior within a single visualization environment.

B. Paper Contributions

This revised paper makes the following concrete contributions:

- 1) it documents a deployable, zero-installation web tool for recursion-tree exploration;
- 2) it formalizes the system as a three-layer architecture with explicit state coordination;
- 3) it presents visual assets and structured tables that were missing from the original report;

TABLE I
COMPARISON OF EXISTING LEARNING RESOURCES AND ALGONEXUS

Tool / Resource	Browser Access	Recursion Focus	Step Annotation	Multi-Algorithm Comparison
SRec [2]	Limited	Strong	Moderate	Low
Online Python Tutor [3]	Yes	General execution	Limited	Low
VisuAlgo [4]	Yes	Broad AV	Moderate	Moderate
Course notes / textbooks [5], [8], [9]	Yes	Conceptual	Static	Low
AlgoNexus	Yes	Strong	Strong	Strong

- 4) it explains how annotation, animation, and color semantics support conceptual learning; and
- 5) it provides an expanded IEEE-style discussion of limitations, extensibility, and pedagogical use.

II. RELATED WORK AND MOTIVATION

Algorithm visualization has long been recognized as useful in computer science education, but the literature consistently shows that the benefit depends on how the visualization is used. The meta-study by Hundhausen *et al.* concluded that active learner engagement is more valuable than passive exposure to animation alone [1]. This principle is directly relevant to recursion teaching, where simply displaying output values rarely helps students build a robust execution model.

Existing educational tools each address part of this need. SRec specifically focused on recursion teaching and remains pedagogically notable, but its deployment model reflects an earlier era of classroom software distribution [2]. Online Python Tutor provides powerful execution tracing and is highly effective for code-level reasoning, yet it is not specialized for recursion-tree analysis across canonical divide-and-conquer patterns [3]. VisuAlgo provides broad algorithm coverage and polished animation, but its emphasis is on general algorithm behavior rather than tightly annotated recursion-tree walkthroughs [4]. Textbook and course resources such as Runestone, MIT OpenCourseWare, and Cornell’s recursion tree notes provide strong conceptual explanations, though they are not themselves dedicated interactive tree visualizers [5], [8], [9].

The practical motivation for AlgoNexus emerges from this gap. Students often need a focused environment in which recursion is treated not as incidental control flow but as the primary object of analysis. For divide-and-conquer topics in a DAA course, an educational tool should ideally be browser-deployable, support multiple algorithms, preserve step-by-step state, and explain why each subproblem is generated. The original report identified exactly these deficiencies, and the revised document makes that gap analysis more explicit.

III. PROBLEM DEFINITION AND DESIGN GOALS

The core problem addressed by the project is representational rather than purely computational. Students are able

to execute recursive algorithms on paper, but they frequently struggle to visualize the expanding call structure that links subproblems, base cases, and returned results. In divide-and-conquer algorithms this difficulty compounds because understanding performance requires connecting the recursion tree with recurrence relations and asymptotic complexity.

From that problem statement, the design goals are straightforward. The system must: 1) depict recursive calls as a tree rather than only as stack frames; 2) preserve execution order so that students can step forward and backward; 3) expose algorithm-specific context such as pivots, search intervals, geometric partitions, or matrix block products; 4) remain easy to run in standard browsers; and 5) provide enough variation in algorithm behavior to support comparative reasoning. These goals guided the module selection, interface layout, and annotation model documented below.

IV. SYSTEM ARCHITECTURE

AlgoNexus follows a modular three-layer architecture that separates interaction, algorithmic computation, and visual rendering. Such separation is useful both educationally and technically: the learner experiences a coherent interface while the implementation remains extensible enough to support new algorithms with minimal redesign.

A. User Interface Layer

The user interface layer contains the algorithm selector, input panel, execution controls, side-panel explanation area, and drawing canvas. Its purpose is not merely data entry. It also shapes how students reason about an algorithm. For example, presenting a sorting input array, a search target, or matrix dimensions directly beside the visual output makes it easier to map symbolic inputs onto recursive state changes. Playback controls such as *execute*, *next*, *previous*, and *reset* transform the tool from a static diagram generator into an exploratory environment.

B. Application Logic Layer

The application logic layer contains the five algorithm engines together with a central *Tree State Manager*. Each engine converts the semantics of its algorithm into a sequence of node-generation events. A node records the current subproblem, depth, parent relation, and a short explanatory annotation. The Tree State Manager stores these nodes in traversal order and exposes an interface for sequential navigation. This layer is the conceptual heart of the system because it bridges formal algorithms and pedagogical representation.

C. Rendering Layer

The rendering layer computes layout positions, draws nodes and edges, and applies animation and color updates. Although the final output appears simple, clear rendering is critical: a cluttered or visually unstable tree would undermine learning rather than support it. The layout strategy is inspired by tidy tree-drawing ideas for hierarchical structures [12]. Vertical position corresponds to recursion depth, while horizontal spacing

Three-layer architecture of AlgoNexus

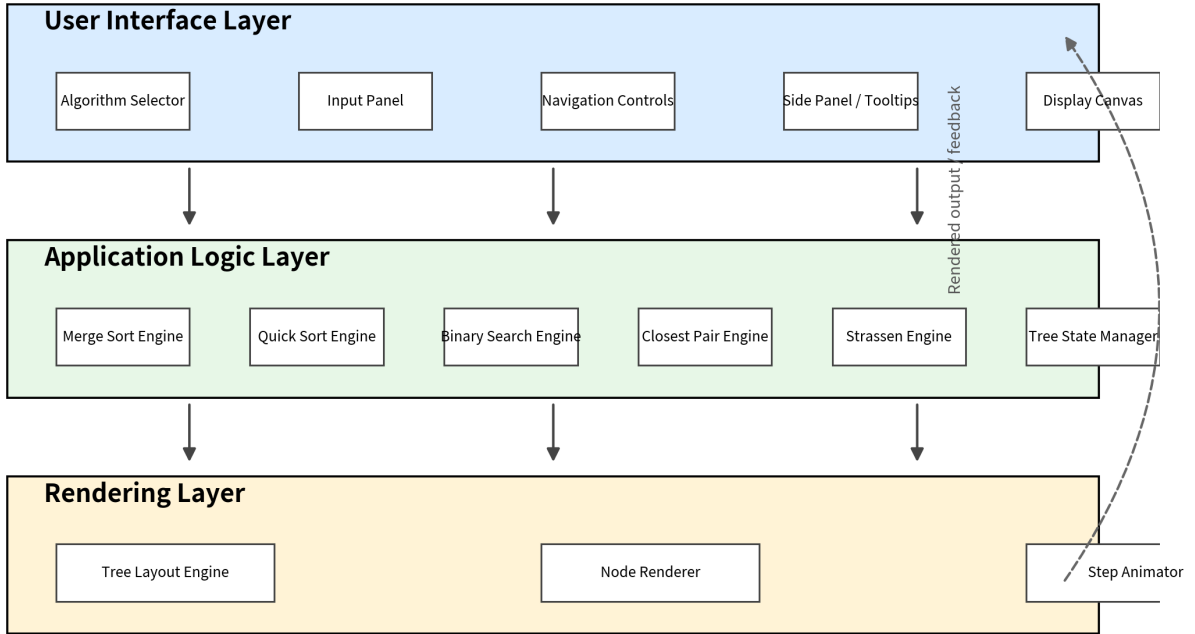


Fig. 1. Three-layer architecture of AlgoNexus. The user interface captures algorithm selection and input, the logic layer generates recursion events and annotations, and the rendering layer transforms these events into a step-wise visual tree.

attempts to preserve subtree balance and reduce overlap. Color coding further encodes state: blue for the currently active node, green for already visited nodes, and gray for pending nodes.

D. Interface Snapshot

Fig. 2 shows the deployed interface on the hosted project site. Even before execution begins, the layout reveals the intended learning workflow: input and algorithm metadata on the left, the evolving recursion tree in the center, and execution trace plus source code on the right. This arrangement allows a student to connect input data, runtime state, and implementation logic without switching contexts.

V. ALGORITHM COVERAGE AND VISUALIZATION MODEL

The project deliberately includes algorithms with distinct recursive signatures. Rather than treating all divide-and-conquer methods as interchangeable examples, the platform highlights the structural differences that shape both intuition and complexity.

A. Why Multiple Algorithms Matter

A single recursion visualizer can help with one topic, but a *comparative* visualizer helps students generalize. Merge Sort and Closest Pair both satisfy an $O(n \log n)$ recurrence yet

differ in domain and combination logic. Binary Search shows that not all recursion trees are full binary trees. Quick Sort demonstrates that identical high-level decomposition rules can still yield very different tree shapes under different pivots. Strassen’s method then extends the discussion by introducing a seven-way split whose complexity exponent is smaller than cubic but substantially different from the other algorithms.

B. Node Model and Annotation Strategy

Every recursive call is represented as a node object containing at least the following fields: a unique identifier, parent identifier, recursion depth, problem-specific parameters, intermediate semantic information, and a short annotation. This annotation field is one of the distinguishing design decisions of the project. Instead of showing only a label such as “left half” or “pivot partition,” the tool describes why the current subproblem exists and how it contributes to the final answer. For educational use, this shifts the visualizer from a raw execution trace toward conceptual explanation.

C. Traversal Semantics

Nodes are stored in depth-first execution order because this order closely mirrors how recursive functions behave at runtime. When a user steps through the process, the active

TABLE II
ALGORITHM COVERAGE IN THE CURRENT PHASE OF ALGONEXUS

Algorithm	Recurrence / Pattern	Typical Complexity	Branching Factor	Pedagogical Value in the Visualizer
Merge Sort	$T(n) = 2T(n/2) + O(n)$	$O(n \log n)$	2	Demonstrates balanced decomposition and explicit merge phase.
Quick Sort	Partition-based recursion	Average $O(n \log n)$, worst $O(n^2)$	Usually 2	Shows that tree shape depends on pivot quality, making imbalance visible.
Binary Search	$T(n) = T(n/2) + O(1)$	$O(\log n)$	1	Illustrates repeated halving and shallow recursion.
Closest Pair of Points	$T(n) = 2T(n/2) + O(n)$	$O(n \log n)$	2	Extends visualization from arrays to geometry and cross-strip reasoning.
Strassen Matrix Multiplication	$T(n) = 7T(n/2) + O(n^2)$	$O(n^{\log_2 7}) \approx O(n^{2.807})$	7	Makes nontrivial branching-factor effects on complexity directly observable.

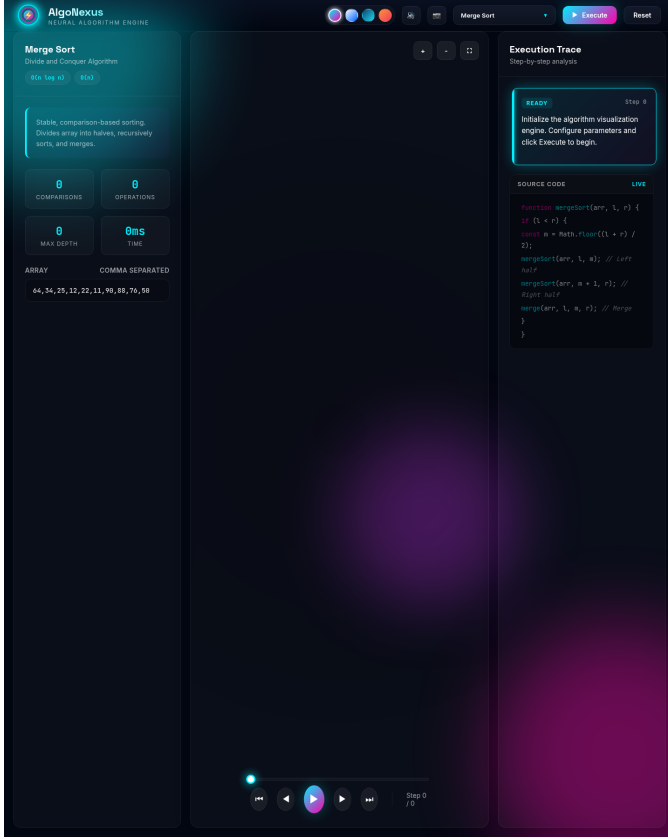


Fig. 2. Screenshot of the deployed AlgoNexus interface showing algorithm metadata, the visualization canvas, playback controls, and execution trace panel.

node is the next node in that depth-first sequence. Backward navigation therefore serves as a lightweight replay mechanism, allowing a student to revisit exactly where a particular sub-problem was introduced.

VI. ILLUSTRATIVE RECURSION STRUCTURES

The current paper adds the figures that were previously described only as placeholders. These visuals strengthen the report because they connect the textual discussion with actual recursion shapes and interface artifacts.

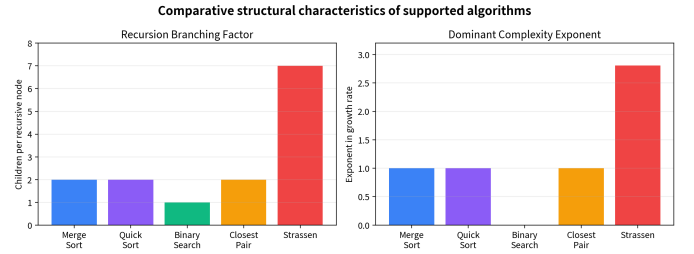


Fig. 3. Structural comparison of the supported algorithms. The visualization tool is particularly useful because it lets students see how branching factor and dominant growth behavior vary across algorithm families.

A. Merge Sort Example

Fig. 4 shows an illustrative Merge Sort recursion tree for a sample input array. The figure emphasizes two key ideas. First, the tree is balanced because the problem is repeatedly divided into near-equal halves. Second, the color semantics allow a learner to distinguish history, current focus, and remaining work at a glance. In the full tool, these states update continuously as the learner advances through execution.

B. Strassen Decomposition Example

Strassen's algorithm is pedagogically important because its advantage is harder to appreciate from equations alone. A student may memorize that the algorithm performs seven recursive multiplications instead of eight, but the recursion tree makes the consequence of that choice more concrete. Fig. 5 depicts the first-level decomposition into products M_1 through M_7 . When this pattern repeats over multiple levels, the rapid growth in the number of nodes also becomes visually apparent, reinforcing the relationship between branching factor and asymptotic behavior.

VII. DETAILED MODULE RESPONSIBILITIES

Table III summarizes the primary modules that collaborate during visualization. The system is intentionally decomposed so that a new algorithm can be integrated by implementing its own engine while reusing the state manager, rendering pipeline, and interaction shell.

Illustrative Merge Sort recursion tree

Blue = active node, green = visited, gray = pending

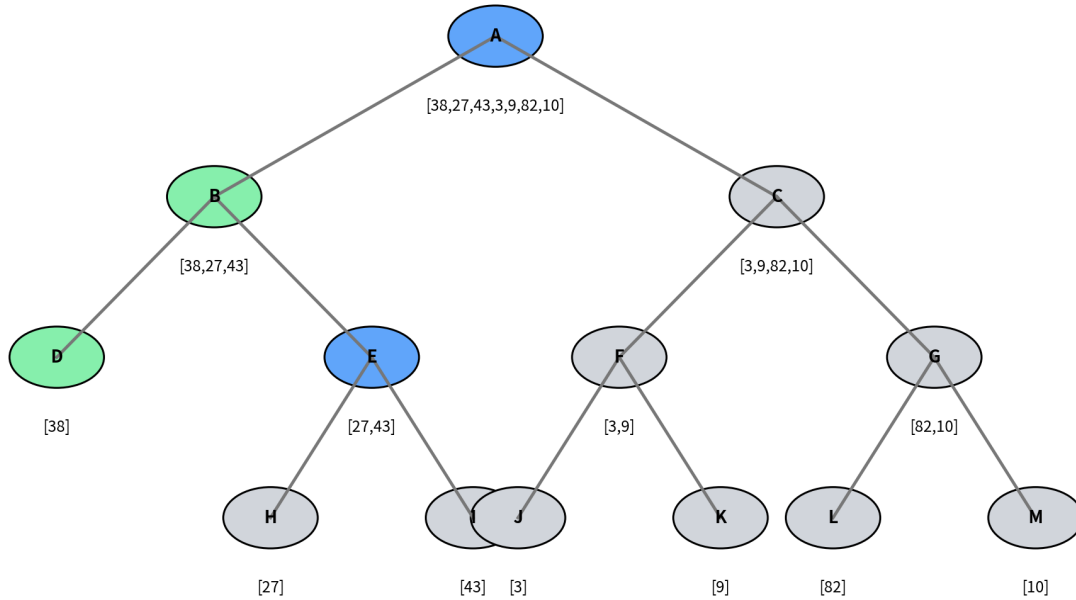


Fig. 4. Illustrative Merge Sort recursion tree showing balanced binary decomposition and the three visual execution states used by the interface.

TABLE III
CORE MODULES IN THE CURRENT IMPLEMENTATION

Module	Layer	Primary Input / Output	Responsibility
Algorithm Selector	UI	User choice → selected engine	Chooses the divide-and-conquer algorithm to visualize and updates input expectations.
Input Panel	UI	Raw user values → validated data	Collects arrays, points, targets, or matrices and passes structured input downstream.
Navigation Controls	UI	Step events	Execute, move forward, move backward, and reset visualization state.
Side Panel / Tooltip View	UI	Current-node annotation	Explains the semantic meaning of the active recursion step.
Algorithm Engines	Logic	Structured input → node events	Generate recursion nodes, intermediate values, and algorithm-specific annotations.
Tree State Manager	Logic	Node events → synchronized state	Maintains traversal order, current step index, and shared visualization state.
Tree Layout Engine	Rendering	Tree structure → coordinates	Computes depth-aware node positions and subtree spacing.
Node Renderer and Animator	Rendering	Coordinates + state → SVG scene	Draws the tree and updates colors/transitions as the active step changes.

VIII. IMPLEMENTATION DISCUSSION

Although the tool is intended for learning rather than high-performance scientific computing, its implementation still requires careful design choices to keep interaction smooth and the display intelligible.

A. Input Handling

The platform does not depend on a fixed external dataset. Instead, it accepts user-defined runtime inputs: integer arrays for sorting, sorted arrays plus targets for searching, coordinate pairs for geometric input, and square matrices for Strassen’s method. This decision is educationally useful because students

can test their own examples, but it also requires validation. Sorting algorithms must reject malformed arrays, Binary Search must enforce sorted input, and Strassen’s method must ensure compatible square matrices, padding dimensions when necessary.

B. State Consistency

The Tree State Manager ensures consistency across navigation and rendering. A common risk in educational visualizers is desynchronization: for example, the trace panel may advance while the diagram does not, or the explanatory text may refer to a different state than the highlighted node. Centralized state management minimizes such errors by ensuring that

First-level 7-way decomposition in Strassen's algorithm

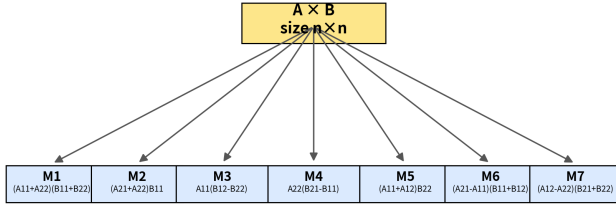


Fig. 5. First-level decomposition used in Strassen's Matrix Multiplication. The visualizer uses this structure to explain why the algorithm replaces eight naive block multiplications with seven recursive products.

the current step index determines every dependent visual component.

C. Rendering Trade-offs

The use of a browser-based graphics pipeline is a deliberate deployment choice. Browser delivery removes installation barriers and improves accessibility for classrooms, demonstrations, and self-study sessions. However, web rendering also introduces practical trade-offs. Large recursion trees can become crowded, especially for algorithms with high branching factors. As input sizes grow, node overlap, text truncation, and panning demands become more pronounced. Thus the current design prioritizes pedagogical clarity over extreme scalability.

D. Source-Code Awareness

An additional educational feature of the interface is the adjacent source-code panel. This enables an important two-way mapping: learners can inspect code and then observe its recursive consequences in the tree, or they can inspect the tree and infer which lines of code produced the current state. Such dual representation is particularly helpful in DAA instruction because it connects implementation details with abstract analysis.

IX. ILLUSTRATIVE ALGORITHM WALKTHROUGHS

Because the visualizer is designed for comparative learning, each supported algorithm is mapped to a characteristic recursion pattern.

A. Merge Sort and Quick Sort

Merge Sort produces one of the cleanest recursion trees in the platform. Learners can see an almost symmetric binary decomposition followed by an implicit combine phase, making it a strong introductory case for recurrence reasoning. Quick Sort complements this by showing that recursion trees are not determined by problem size alone; they also depend on the pivot and the resulting partition balance. In classroom use, these two algorithms together make it easy to discuss why two algorithms may share an average $O(n \log n)$ complexity while still producing visually different execution histories.

B. Binary Search

Binary Search is useful precisely because its tree is not a broad tree at all. The visualization effectively collapses into a single descending branch, allowing students to connect interval halving with logarithmic depth. This is particularly helpful for learners who initially interpret all recursive algorithms as having binary expansion. The visualizer corrects that misconception immediately.

C. Closest Pair of Points

The Closest Pair module expands the educational scope beyond arrays. Students observe that divide-and-conquer also applies to geometric problems in which the combine stage must reason about cross-boundary candidates. This module therefore encourages transfer of understanding from familiar list-based examples to computational geometry.

D. Strassen's Matrix Multiplication

Strassen's module serves as the most advanced case in the current phase. It introduces block partitioning, matrix-expression annotations, and a branching factor of seven. For many students, the key insight is not the exact formulas for M_1 through M_7 but the idea that reducing the number of recursive subproblems changes the dominant exponent. The visual tree provides an intuitive route into that observation before formal complexity analysis is attempted.

X. ANALYTICAL EVALUATION AND EDUCATIONAL VALUE

The present phase of the project is best understood as a design-and-demonstration system rather than a completed empirical study. Accordingly, the evaluation in this paper is analytical rather than statistical. The question is not whether the tool has already proven a quantified learning gain in a classroom trial, but whether the system design plausibly supports the learning objectives it targets.

Three observations follow from the current implementation. First, the platform makes structural contrasts between algorithms immediately visible. Binary Search appears as a thin single-branch path, Merge Sort as a balanced binary expansion, and Strassen's method as a higher-fanout decomposition. This direct contrast supports conceptual comparison in ways that static lecture notes usually cannot. Second, contextual annotation reduces the risk that learners will interpret the tree as merely a decorative output. By pairing each node with an explanation, the system foregrounds algorithmic meaning. Third, interactive navigation supports active engagement, which the algorithm-visualization literature consistently associates with better pedagogical outcomes than passive viewing [1].

At the same time, analytical evaluation must remain honest about its boundaries. The project does not yet report controlled classroom measurements, pre-/post-test score changes, eye-tracking evidence, or longitudinal use data. Those forms of evidence belong to a later phase. Still, as a first-phase engineering and educational prototype, the tool demonstrates a coherent alignment between problem statement, design decisions, and expected instructional value.

TABLE IV
KEY DESIGN DECISIONS AND THEIR INTENDED LEARNING EFFECT

Design Element	Intended Educational Effect
Step-wise navigation	Supports self-paced exploration and revisiting of confusing recursion states.
Annotation side panel	Explains why a subproblem exists rather than only what data it contains.
Color-coded nodes	Distinguishes active, completed, and pending calls without requiring textual interpretation.
Multiple algorithms	Enables comparison of recursion depth, balance, and branching structure.
Browser deployment	Reduces setup friction for classroom and home use.
Source-code panel	Links implementation statements to visual runtime behavior.

XI. LIMITATIONS AND FUTURE WORK

The project already offers meaningful instructional functionality, but its current form has several limitations. First, algorithm coverage, although broader than many recursion-specific tools, remains incomplete relative to a full DAA curriculum. Natural extensions include Tower of Hanoi, Fibonacci recursion, Karatsuba multiplication, and additional computational-geometry algorithms. Second, the present report does not include controlled empirical evaluation with students. Without such evidence, claims about learning impact must remain theoretically motivated rather than experimentally confirmed.

Third, very large trees can exceed comfortable screen density, especially for Strassen’s algorithm. This creates a practical need for adaptive zooming, panning, collapsing, or focus-plus-context strategies. Fourth, persistent session state is limited; a browser refresh may interrupt the learning workflow. Fifth, although the current annotations are useful, richer explanatory layers could be added in the future, including recurrence display, Master Theorem hints, and prompts that ask the learner to predict the next recursive step.

Future work should therefore proceed on three parallel tracks: pedagogical validation, interface scalability, and algorithmic expansion. A well-designed classroom study could compare learning outcomes before and after exposure to the visualizer. In parallel, the interface could incorporate saved sessions, larger-input navigation aids, and adaptive labeling. Finally, the architecture is already modular enough to support new algorithm engines without restructuring the entire application, making the system suitable for iterative course-project growth.

XII. DEPLOYMENT AND ACADEMIC USE

The current deployment on GitHub Pages is a practical advantage because it makes the project immediately shareable during lab demonstrations, peer review, and course evaluation. A student or instructor can open the interface without installing runtime dependencies, meaning the attention remains on algorithm analysis rather than on setup. This low-friction deployment also supports iterative project growth: the hosted

site can be updated as new algorithms, figures, or explanatory features are added.

From an academic-reporting perspective, the project is also suitable for future extension into a more formal software-engineering or educational-technology study. The system already has identifiable modules, a reproducible public deployment, and a bounded pedagogical objective. These properties make it a strong candidate for a second-phase report that includes usability testing, comparative studies against existing teaching aids, and more rigorous measurement of conceptual understanding.

XIII. CONCLUSION

This revised IEEE-format report presents AlgoNexus as a focused educational system for understanding divide-and-conquer recursion through interactive tree visualization. The project addresses a specific instructional problem: learners often know how to read recursive pseudocode and solve recurrences, yet still struggle to picture the evolving call structure that gives those formulas meaning. By combining browser accessibility, structured interaction, contextual annotation, and multiple representative algorithms, the tool makes recursion more concrete, comparable, and discussable.

Relative to the original report, the present version provides a stronger academic presentation through formal organization, added figures, richer comparative discussion, and clearer articulation of design rationale. The project remains a first-phase implementation, but it already demonstrates how a targeted visualization system can complement conventional lectures, notes, and textbook examples. With empirical validation and further feature expansion, AlgoNexus has the potential to become a valuable companion resource for DAA instruction focused on recursion and divide-and-conquer analysis.

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